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The mechanisms of calcium mobilization by procyanidins, flavonols and flavonoids from *Cecropia glaziovii* Sneth in pulmonary endothelial cell cultures endorse its popular use as vasodilator phytomedicine

Gabriella Trettel [a,c,1], Clelia Rejane Antonio Bertoncini [b,c,*], Maria Teresa Lima-Landman [a]

[a] *Department of Pharmacology, Federal University of São Paulo, Brazil*
[b] *CEDEME, Centre of Development of Experimental Models for Medicine and Biology, Federal University of São Paulo, Brazil*
[c] *Department of Gynaecology, Federal University of São Paulo*

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ABSTRACT

The hypotensive and antihypertensive activities of the aqueous extract (AE) and butanolic fraction (ButF) isolated from *Cecropia glaziovii* Sneth have been demonstrated in previous studies in animal models. This study aimed to evaluate the molecular mechanism of action responsible for the vasodilatory effect of procyanidins, flavanols, and flavonoids found in *C. glaziovii* in endothelial cell culture. For this purpose, we analyzed the effect of procyanidin B2 and B3 compounds, catechin, epicatechin, orientin, isoorientin, and isovitexin in the mobilization of Ca^{2+} in rat endothelial cell cultures. Parallel associations with different antagonists were examined by considering the following in vivo hypotensive mechanisms: blockage of L-type calcium channels, action on β-2 adrenergic receptors, and vasodilation via the nitric oxide pathway. All measurements of calcium mobilization were carried out by using the fluorescence measurement methodology in a Flexstation M3 spectrophotometer. The results indicate that some of the compounds have mixed actions, acting through different calcium mobilization pathways. The mobilization induced by such compounds significantly decreased when they were incubated with their corresponding antagonists. Taken together, our data suggest that the beneficial effects seen with the popular use of *Cecropia glaziovii* Sneth in pathological conditions, such as systemic arterial hypertension, seem to be related to the plant's hypotensive effect, very probably promoted by the actions of flavonols, flavonoids, and procyanidins, by different pathways of calcium mobilization.

1. Introduction

Cecropias are trees characterized by rapid growth and appear naturally in deforested areas, clearings, forest edges, or in humid places such as swamps and floodplains and are popularly used for the treatment of respiratory tract (cough, pneumonia, asthma, bronchitis), heart and circulatory diseases, diabetes, hepatitis, inflammation, and ulcers [1,2].

Cecropia glaziovii Sneth is typical of tropical and subtropical regions of Latin America and is found in central and southern Brazil [3].

Previous studies from our group have demonstrated the hypotensive and antihypertensive activities of the aqueous extract (AE) and the butanolic fraction (ButF) isolated from *Cecropia glaziovii* Sneth. Due to biomonitored isolation, a hypotensive action of the plant's AE was maintained in its ButF, but in which procyanidins, catechins, and flavonoids were the major constituents [2]. While the antidepressant activity of ButF was measured in rat models [2], the anxiolytic effect of AE was observed in mice [4].

Hypotensive activity occurs when the relaxation of smooth muscles takes place, which can be stimulated by several mechanisms [5–15] such as the following: blockage of L-type calcium channels [7–20], action on β-2 adrenergic receptors [21], and/or participation of the nitric oxide pathway [20–23].

Previous studies carried out in our laboratory have demonstrated the hypotensive and antihypertensive activities of the aqueous extract (AE)

Abbreviations: ACh, acetylcholine; ANG II, Angiotensin II; ATR, Atropine; AT1, Angiotensin II receptor subtype 1; BK, Bradykinin; ButF, butanolic fraction; DMEM, Eagle medium modified by Dulbecco; AE, aqueous extract; EDTA, Ethylene-diamine tetra acetic acid; FBS, fetal bovine serum; GMPc, Guanosine cyclic monophosphate; KCL, potassium chloride; NO, nitric oxide; NOS, nitric oxide synthase.

* Corresponding author at: CEDEME, Centre of Development of Experimental Models for Medicine and Biology, Federal University of São Paulo, Brazil.
E-mail addresses: gabriella.trettel@unifesp.br (G. Trettel), clelia.rejane@unifesp.br, cleliabertoncini@hotmail.com (C.R.A. Bertoncini).
[1] Present address: Department of Gynaecology, Federal University of São Paulo.

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and the butanolic fraction (ButF) isolated from *Cecropia glaziovii* Sneth. In 1996, Cysneiros analyzed the hypotensive activity of the aqueous extract from the leaves of *C. glaziovii*. The extract showed hypotensive activity promoting smooth muscle relaxation, by blocking L-type calcium channel mechanisms and by action on β-2 adrenergic receptors, according to in vivo experiments [21].

The hypotensive action of the AE of the plant and its Fbut was also confirmed in other studies carried out by the group, mentioning the hypotensive and antihypertensive activities of AE and ButF both in normotensive rats and in hypertensive rats, after chronic oral administration (O.V – gavage). The in vivo evaluation of the hypotensive activity of the aqueous extract of *C. glaziovii* demonstrated that a single administration of the extract did not cause changes in blood pressure or heart rate. However, repeated administrations reduced blood pressure after 14 days of treatment. The hypotension caused by the plant extract is related to the blocking of smooth muscle Ca2 + channels [31].

Previous studies suggest that the intake of flavanols, such as epicatechin and catechins, is associated with reduced risk of cardiovascular diseases due to their vasodilating properties [24]. Catechin and its oligomers, procyanidins, also belong to the group of flavanols [25]. Cardiovascular protective effect, as revealed through the analysis of vascular vasodilation, indicated a possible decrease in blood pressure when individuals were eating foods rich in flavonols [26].

For the therapeutic control of arterial hypertension, researchers use different types of drugs, including calcium channel blockers, especially verapamil. Verapamil acts on voltage-gated calcium channels inhibiting Ca^{2+} ion entry and antagonizing the function of these channels, which results in muscle relaxation with a consequent decrease in peripheral vascular resistance and blood pressure. The use of vasoconstrictor and vasodilator drugs allows the evaluation of the functioning of cells in culture against different agonists and antagonists with known and different mechanisms of action [30].

Animal experiments have shown that epicatechin is capable of preventing and reducing hypertension by improving endothelial function, reducing oxidative stress in cardiovascular diseases, and inflammatory state, the first events that mediate the development of atherosclerosis and insulin resistance [27].

In vitro, epicatechin is capable of inducing endothelium-dependent vasodilation, mediated by the nitric oxide pathway. The vasorelaxant effect of procyanidins is induced by increased release of NO in the vascular endothelium and studies suggest that this increase is due to the activation of NOS, with an increase and improvement in endothelial function from the increase in intracellular calcium concentration [28].

Therefore, in addition to increasing NOS activation, the endothelium-derived hyperpolarizing factor (EDHF) is released. EDHF is an important factor that participates in the regulation of homeostasis and induces vasodilation. However, the exact mechanism of action of the EDHF is not fully understood, despite the indisputable regulation of endothelial exercise by it [29].

C. glaziovii Sneth is composed predominantly of metabolites of phenolic acids, catechins, procyanidins, and flavonoids. Using the high-performance liquid chromatography (HPLC) method to analyze the methanolic extract of its scales located on the stem (stipules), the procyanidin compounds B2, B3, and C1, epicatechin, and catechin were identified. The presence of chlorogenic acid and also of the procyanidins B2 and C1 and epicatechin were identified in the methanolic extract of the leaves. As a major constituent, flavonoids, especially isoorientin, were identified [23,32–34]. However, in the project, all compounds were obtained from Sigma Aldrich.

Based on the above, this study aimed to analyze intracellular calcium mobilization in pulmonary endothelial cells, stimulated/blocked by compounds present in *C. glaziovii* Sneth (isovitexin, isoorientin, catechin, epicatechin, procyanidin B2, procyanidin B3, and orientin) in combination with antagonistic drugs. Thus, we seek to further elucidate the mechanisms involved in the hypotensive/antihypertensive action of the substances derived from *C. glaziovii* Sneth in intracellular calcium homeostasis.

2. Materials and methods

2.1. Animals

50 male Wistar-INFAR rats, approximately 3 months old, normotensive, from the Instituto Nacional de Farmacologia (INFAR), Escola Paulista de Medicina (EPM)/Universidade Federal de São Paulo (UNIFESP), were used in this study. The animals were kept under controlled conditions of temperature (20–25 °C) and artificial lighting (12 h light/12 h dark cycle) with free access to water and food. All the experimental protocols were approved by the Ethics Committee on the Use of Animals of UNIFESP (CEUA: 7742150916).

2.2. Drugs

Angiotensin II, antibiotics streptomycin and penicillin (1%), atropine, bradykinin, acetylcholine chloride, compounds isovitexin, isoorientin, catechin, epicatechin, procyanidin B2, procyanidin B3, and orientin, eagle medium modified by Dulbecco – DMEM (Sigma-Aldrich, USA), fetal bovine serum – FBS (Gibco Life Technologies, USA), HOE 140 (Sigma Aldrich), loading buffer (Fura-4) + probenecid + HBSS (FLIPR® Calcium-4 assay kit- Molecular Devices), losartan potassium (Fluka), phosphate buffer solution (PBS), trypsin-EDTA 1 × 0,05% (Gibco, USA), trypan blue (Sigma-Aldrich).

2.3. Experimental protocols

2.3.1. Primary culture of endothelial cells from pulmonary microvasculature

The animals were euthanized by decapitation, the lungs were removed and washed with phosphate buffer solution (PBS). The tissue from three animals per group was chopped into pieces of approximately 1 mm^3 and placed in 10 mL plastic bottles, with Dulbecco's Modified Eagle Medium (DMEM) supplemented with 20% fetal bovine serum (FBS) and antibiotics (streptomycin and penicillin 1%) at pH 7.4 and kept in an incubator at 37° C, 5% CO^2, for 60 h. After this period, the supernatant tissue was discarded and the medium was changed to eliminate red blood cells [4]. The changes in the nutrient medium took place every 2–3 days. When the cell confluence reached about 80% of the plate, the cells were subcultured using 0.05% EDTA trypsin (GIBCO) and transferred to 20 mL bottles [5].

The viability of endothelial cells against vasoconstrictor and vasodilator drugs has been standardized to determine the appropriate concentration of agonists and antagonists with known mechanisms of action, cell density, and the number of (subcultural) passages required to measure intracellular calcium mobilization. (Results in supplementary material).

Pulmonary endothelial cell cultures were used between the third and fifth subcultures. After the endothelial cells showed 80% confluence, they were subcultured until the desired number of subcultural passages (between the third and fifth) required to measure intracellular calcium mobilization was obtained. For testing, the cells were transferred to 96-well microplates, and this step of the procedure was performed in duplicate of culture wells per treatment.

2.3.2. Measurement of fluorimetry-free cytosolic calcium in endothelial cells

The cells were plated in duplicate 96-well microplates 24 h before the experiment at densities of 20,000, 40,000, and 80,000 cells per well, and incubated at 37 °C. After 24 h, the culture medium was removed and 50 μL of the mixture was added: loading buffer (Fura-4) + probenecid + HBSS (FLIPR® Calcium-4 assay kit- Molecular Devices), as described in the manufacturer's instructions. The plates were incubated for 1 h before reading, and when the blockers were used, they were added after

50 min of incubation. The experiments were conducted in duplicate and the plates were read in a FlexStation® plate reader (Molecular Devices), with λex 494 nm and λem 525 nm. The originated fluorescence that represents the cytosolic free calcium concentration was evaluated as ΔUAF, which corresponds to the ratio of the maximum fluorescence point minus the basal fluorescence:

$$\Delta UAF \quad MAX - \Delta MIN$$

All the results obtained were extracted from the fluorescence curve after 90 s of the addition of the tested substances [2,5,21,23,31].

2.3.3. Statistic analyses

The results were expressed as mean ± standard error of the mean. The comparison between the two groups was performed by Student's *t*-test, with $p < 0,05$. When comparing more than one variable, the 'ANOVA' test was performed followed by the 'Tukey' post hoc test, which was marked by an asterisk for a significant difference $p < 0.05$. All results were analyzed using GraphPad Prism Software (San Diego – CA, version 5.0).

3. Results

3.1. Standardization of primary endothelial cell culture with agonist and antagonist drugs

We standardize endothelial cells against agonists (KCl, angiotensin II, bradykinin and acetylcholine, Figs. A–D) and their respective antagonists (verapamil, losartan, HOE 140 and atropine, Figs. E–H) testing cell viability, as presented in the supplementary material.

3.2. Measurement of calcium mobilization by the compounds presents in C. glaziovii Sneth

In the presence of isovitexin (3, 10, 30, and 100 μM), Δ UAF was increased by 2, 2.5, 4 and 11 times, respectively, in relation to HBSS (Δ UAF = 2.1 ± 0.36 n = 3). Isoorientin (3, 10, 30, and 100 μM) increased the fluorescence generated by 2 and 3 times only with the two highest concentrations tested (30 and 100 μM, respectively) (Fig. 1).

Catechin (3, 10, 30, and 100 μM) increased the intracellular calcium concentration by 2 times in the highest concentration tested, compared to HBSS (Δ UAF = 2.41 ± 0.30 n = 3). Epicatechin, in turn, showed a concentration-dependent response, increasing cytosolic calcium by 1, 1.5, 2, and 6 times, at concentrations of 3, 10, 30, and 100 μM, in relation to HBSS, respectively (Fig. 2).

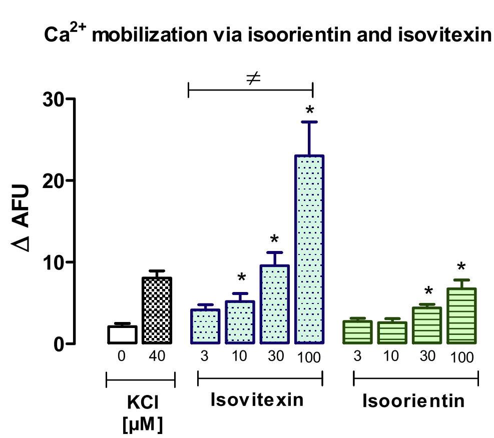

Fig. 1. Free cytosolic Ca2 + mobilization induced by isoorientin and isovitexin (3, 10, 30 and 100 μM) in lung endothelial cell culture (third pass) at a density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), with a significant difference p < 0.05 (*) in relation to HBSS (KCl 0) and ≠ in relation to 3 μM isovitexin.

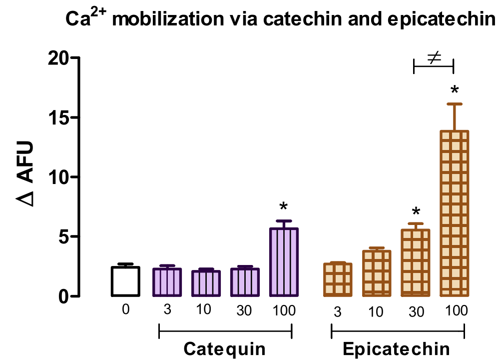

Fig. 2. Free cytosolic Ca2 + mobilization induced by catechin e epicatechin (3; 10; 30 e 100 μM) in lung endothelial cell culture (third pass) at a density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), with a significant difference p < 0.05 (*) in relation to HBSS (0) and ≠ in relation to 30 μM epicatechin.

In the presence of procyanidin B2, cytosolic calcium was mobilized 2 and 4 times at concentrations of 30 and 100 μM, respectively, in relation to HBSS (Δ UAF = 1.3 ± 0.17, n = 3 animals, n) (Fig. 3). Procyanidin B3 showed a concentration-dependent response to calcium mobilization, whose values were 1.5, 2, 3, and 8 times higher at concentrations of 3, 10, 30, and 100 μM, respectively, compared to HBSS. Orientin showed values of 1.5–3 times in concentrations of 3, 10, 30, and 100 μM, in relation to HBSS (Fig. 3).

3.3. Measurement of calcium mobilization by the compounds present in C. glaziovii against antagonists

All compounds, except isoorientin, showed calcium mobilization through the same pathway of action as the KCl agonist, where the response to verapamil reduced this mobilization (Figs. 4–6). The same response can be seen against the losartan antagonist (Figs. 7–9).

The compounds isovitexin, epicatechin, and procyanidin B3 show reduced calcium mobilization against the atropine antagonist (Figs. 10–12). The compounds isovitexin, isoorientin, epicatechin, and procyanidin B3 also reduced the mobilization of calcium when incubated with HOE 140 (Figs. 13–15).

4. Discussion

Previous studies described the hypotensive and antihypertensive activities of the AE and the ButF of *C. glaziovii* in rats [2]. The

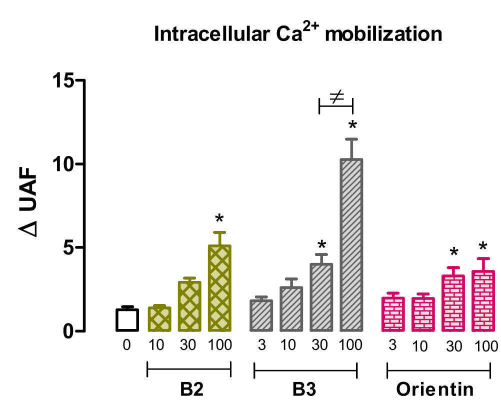

Fig. 3. Free cytosolic Ca2 + mobilization induced by procyanidin B2, procyanidin B3 and orientin (3; 10; 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at a density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), with a significant difference p < 0.05 (*) in relation to HBSS (0) and ≠ in relation to procyanidin B3 30 μM.

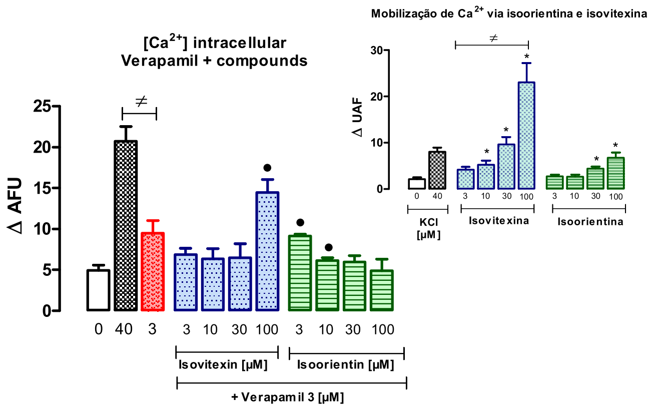

Fig. 4. Free cytosolic Ca2 + mobilization resulting from the combination of the 3 μM verapamil antagonist and the isovitexin and isoorientin compounds (3, 10, 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at a density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), with a significant difference $p < 0.05$ (•) in relation to the pure compounds and ≠ in relation to the 40 μM KCl.

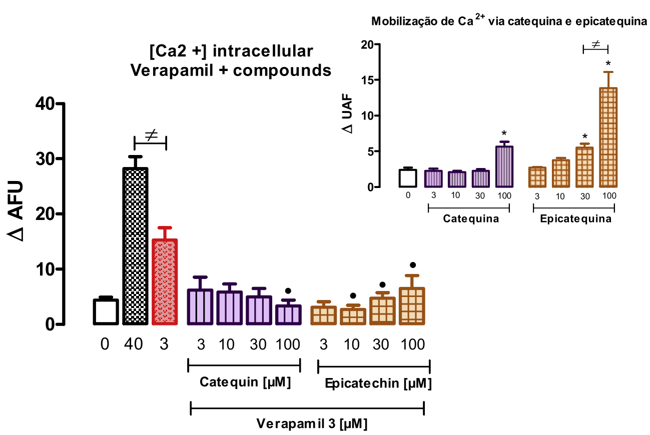

Fig. 5. Free cytosolic Ca2 + mobilization resulting from the combination of the 3 μM verapamil antagonist and the catechin and epicatechin compounds (3, 10, 30 and 100 μM) in lung endothelial cell culture (third pass) at a density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), with a significant difference $p < 0.05$ (•) in relation to the pure compounds and ≠ in relation to the 40 μM KCl.

hypotensive and antihypertensive actions of *C. glaziovii* were related to the presence of flavonoids, flavanols, and procyanidins in their chemical composition.

Here, we aimed to study the mechanisms of the molecular action of procyanidins B2 and B3, catechin, epicatechin, orientin, isoorientin, and isovitexin in cultures of endothelial cells by analyses of intracellular calcium homeostasis.

The intake of polyphenolic compounds is associated with a reduction in the risk of cardiovascular disease due to their vasodilating properties, increasing endothelial activity by increasing calcium, activating, and increasing the nitric oxide responsible for vasodilation and lowering blood pressure. However, the mechanisms by which these processes occur are not yet fully understood [24,31,35–37].

KCl acts in a non-specific way in several cell types, mobilizing intracellular calcium in response to the depolarization of the cell membrane, thus allowing to evaluate the cell's functionality. This response is mediated by potassium channels of the voltage-dependent types and activated by Ca^{2+} (KCa) [38] and are activated when they reach the membrane potential between −35 mV and −55 mV [39].

The renin-angiotensin-aldosterone system (RAAS) is composed of a

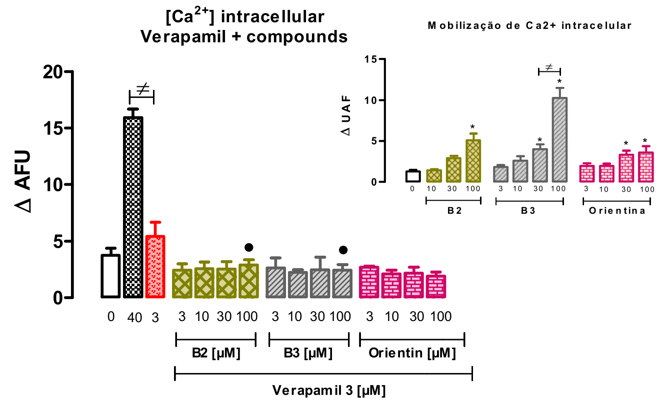

Fig. 6. Free cytosolic Ca2 + mobilization resulting from the combination of the antagonist verapamil 3 μM and the compounds procyanidin B2, procyanidin B3 and orientin (3, 10, 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at the density of 40 thousand cells per well. Data expressed as mean 2 sem (n = 3, in duplicate), with significant difference $p < 0.05$ (•) in relation to pure compounds and ≠ in relation to KCl 40 μM.

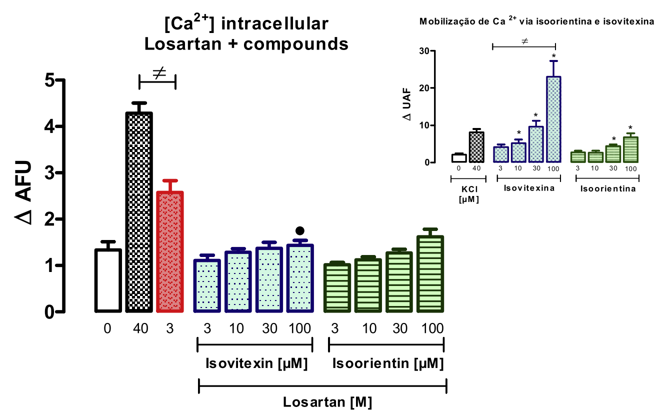

Fig. 7. Free cytosolic Ca2 + mobilization resulting from the association of losartan and the isovitexin and isoorientin compounds (3; 10; 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at a density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), with a significant difference $p < 0.05$ (•) in relation to pure compounds and ≠ in relation to KCl.

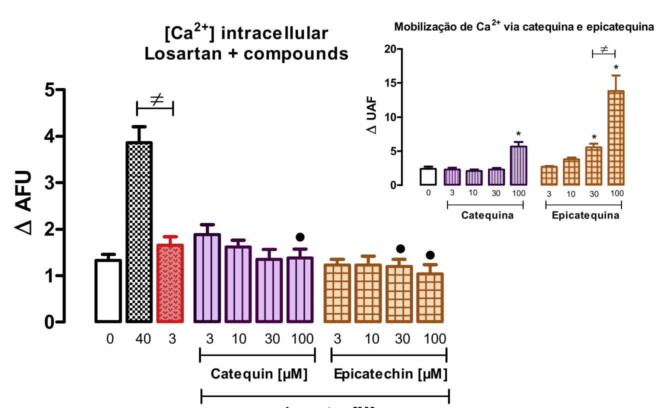

Fig. 8. Free cytosolic Ca2 + mobilization resulting from the association of losartan and catechin and epicatechin compounds (3, 10, 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at a density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), being considered a significant difference $p < 0.05$ (•) in relation to pure compounds and ≠ in relation to KCl.

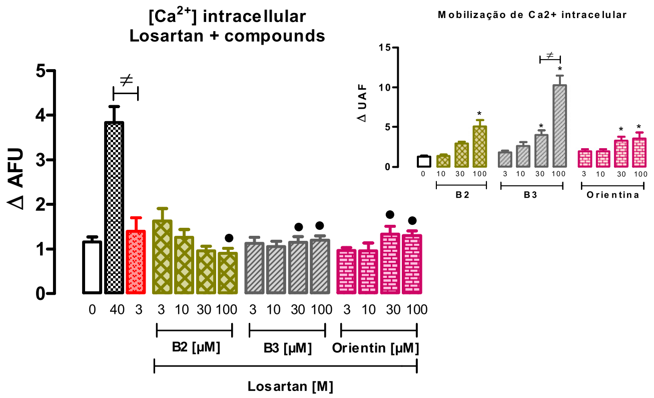

Fig. 9. Free cytosolic Ca2 + mobilization resulting from the association of losartan and the compounds procyanidin B2, procyanidin B3 and orientin (3; 10; 30 and 100 μM) in pulmonary endothelial cell culture 3, (third pass) at the density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), with a significant difference p < 0.05 (•) in relation to pure compounds and ≠ in relation to KCl.

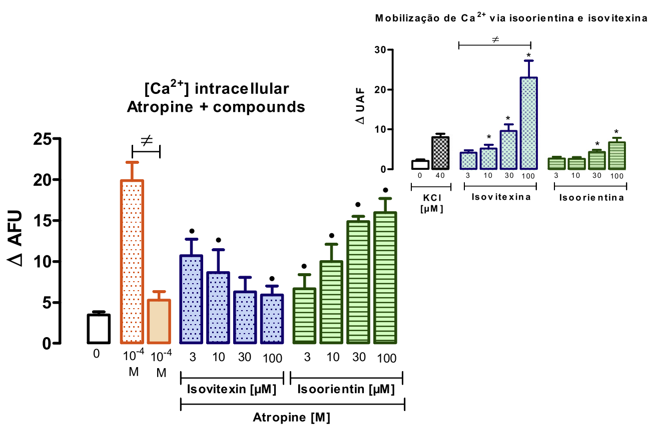

Fig. 10. Mobilization of free cytosolic Ca2 + resulting from the association of atropine and the compounds isovitexin and isoorientin (3, 10, 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at the density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), with a significant difference p < 0.05 (•) in relation to pure compounds and ≠ in relation to ACh.

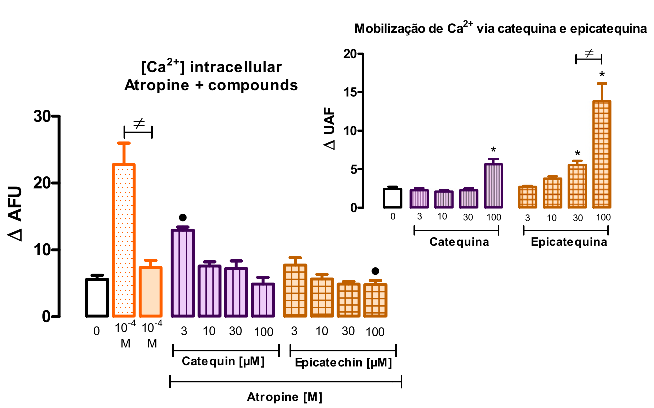

Fig. 11. Free cytosolic Ca2 + mobilization resulting from the association of atropine and the catechin and epicatechin compounds (3, 10, 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at a density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), with p < 0.05 (•) being considered significant difference in relation to pure compounds and ≠ in relation to ACh.

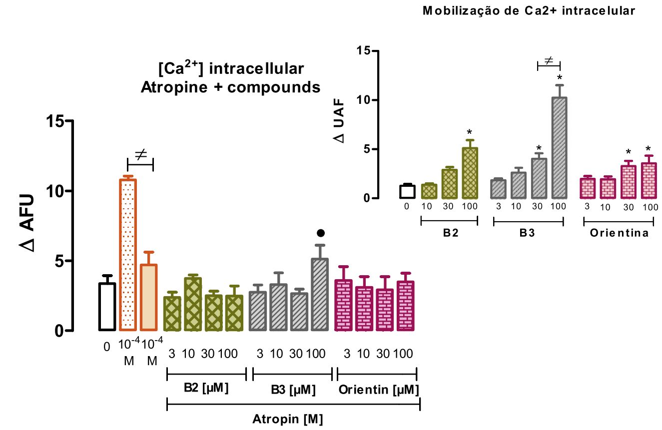

Fig. 12. Free cytosolic Ca2 + mobilization resulting from the association of atropine and the compounds procyanidin B2, procyanidin B3 and orientin (3; 10; 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at the density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), with a significant difference p < 0.05 (•) in relation to pure compounds and ≠ in relation to ACh.

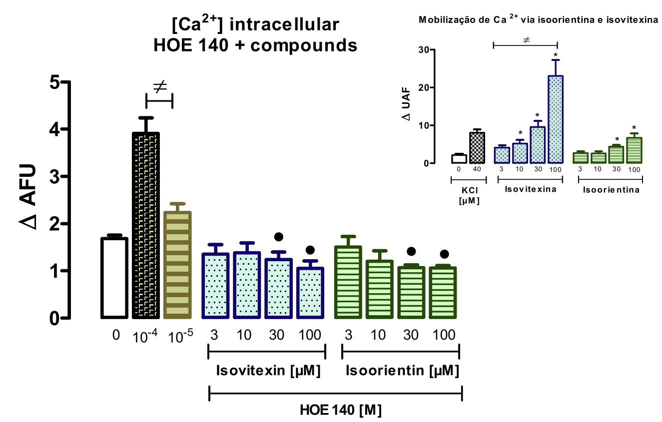

Fig. 13. Free cytosolic Ca2 + mobilization resulting from the association of HOE 140 and the isovitexin and isoorientin compounds (3; 10; 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at a density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), being considered a significant difference p < 0.05 (•) in relation to pure compounds and ≠ in relation to BK.

complex network that regulates blood pressure by the expression of genes and proteins, having a fundamental role in the pathophysiology of SAH, relating to salt and water homeostasis and control of vascular tone [40]. This system consists of the proteins renin (REN), angiotensinogen (AGT), angiotensin-converting enzyme (ACE), and receptors for angiotensin II (AT1 AND AT2) [40,41]. The physiological effects of ANG II are mediated by the AT1 and AT2 receptors, which perform pathological and protective actions, respectively [42–44].

The stimulation of these receptors mobilizes calcium by different mechanisms: AT1 receptors promote vasoconstriction via protein Gq, while AT2 receptors are characterized by vasodilatory properties [45]. When evaluating the compounds isovitexin, isoorientin, catechin, epicatechin, procyanidin B2, procyanidin B3, and pure orientin in endothelial cells, we observed the same concentration-dependent pattern for all. Isovitexin and procyanidin B3, both at concentrations of 100 μM, showed the highest calcium mobilization with levels close to the mobilization seen by KCl, used as a control in these tests.

The vasorelaxant effect of procyanidins is induced by the increased release of NO in the vascular endothelium and studies suggest that this increase is due to the activation of nitric oxide synthase (NOS), with an

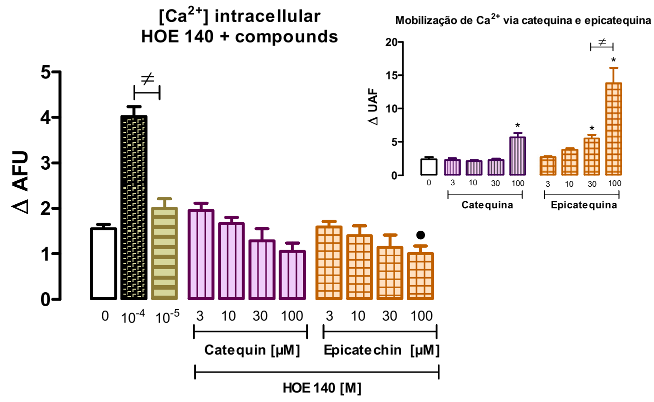

Fig. 14. Free cytosolic Ca2 + mobilization resulting from the association of HOE 140 and the catechin and epicatechin compounds (3, 10, 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at a density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), being considered a significant difference $p < 0.05$ (•) in relation to pure compounds and ≠ in relation to BK.

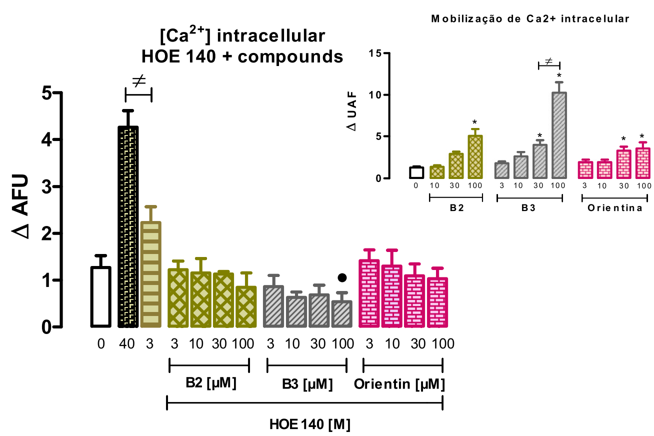

Fig. 15. Free cytosolic Ca2 + mobilization resulting from the association of HOE 140 and the compounds procyanidin B2, procyanidin B3 and orientin (3; 10; 30 and 100 μM) in pulmonary endothelial cell culture (third pass) at the density of 40 thousand cells per well. Data expressed as mean ± sem (n = 3, in duplicate), being considered a significant difference $p < 0.05$ (•) in relation to pure compounds and ≠ in relation to BK.

increase and improvement in endothelial function from the increase in intracellular calcium concentration [46]. In addition to increasing NOS activation, the release of the endothelium-derived hyperpolarizing factor (EDHF) is also stimulated.

Therefore, our study suggests that the mobilization of calcium promoted by the studied compounds of Cecropia is related to vasodilation and stimulation of nitric oxide since these compounds are strongly related to the vasodilator and hypotensive action. However, according to the results presented in Table 1, the compounds have a mixed effect, acting through other channels and not all compounds are related to the studied action.

The greatest mobilization of calcium by these cells occurred at a concentration of 40 μM KCl, regardless of the density of cells used (20, 40, or 80 thousand/well) (Fig. A in supplementary material). Different types of drugs are used for the therapeutic control of arterial hypertension, including calcium channel blockers, especially verapamil. Verapamil acts on voltage-gated calcium channels by inhibiting Ca^{2+} ion entry and antagonizing the function of these channels, which results in muscle relaxation with a consequent decrease in peripheral vascular resistance and blood pressure.

As the concentration of verapamil is increased, the increase in the

Table 1
Effect of the compounds of *C. glaziovii* Sneth in association with the vasodilating antagonists verapamil and losartan and the vasoconstrictors atropine and HOE 140 in pulmonary endothelial cells.

	Effect via calcium channel	Muscarinic effect	Effect AT1 receptors	Effect B2 receptors
Isovitexin	✓	✓	✓	✓
Isoorientin	–	–	–	✓
Catechin	✓	–	✓	–
Epicatechin	✓	✓	✓	✓
Procyanidin B2	✓	–	✓	–
Procyanidin B3	✓	✓	✓	✓
Orientin	✓	–	✓	–

Ca^{2+} concentration promoted by the 40 μM KCl is decreased (Fig. E in supplementary material). Verapamil is known to be a selective drug that specifically inhibits the calcium current through L-type channels, with little effect on the concentrations of sodium (Na^{+}) and magnesium (Mg^{2+}) [6,7]. The contraction/vasoconstriction process can be triggered by several substances, such as angiotensin II, according to the vasoconstrictor drug used in this study. ANG II increases the concentration of intracellular Ca^{2+} in endothelial cells, via the AT1 receptor.

The results obtained in endothelial cells under the action of the vasoconstrictor peptide angiotensin II demonstrated a concentration-dependent response (Fig. B in supplementary material) [8]. Angiotensin II AT1 receptor antagonists are a class of drugs widely used in the treatment of hypertension. Losartan is the AT1 receptor antagonist used in this study. In endothelial cells, ANG II showed a concentration-dependent response, which was blocked when incubated with losartan (Fig. F in supplementary material).

We also used bradykinin in this study, a potent vasodilator with a well-known mechanism: it acts via type B2 receptors present in the vascular endothelium. BK was evaluated in pulmonary endothelial cells with a higher concentration-dependent response among all the agonists used (Fig. C in supplementary material). This fact can be related to the cell type evaluated since the action of BK is dependent on the endothelium [9,10].

Subtype B2 receptors are expressed in most tissues and are significantly present in vascular smooth muscle cells, endothelial cells, and fibroblasts. BK activates the NO cascade via B2 receptors present in endothelial cells, which promotes vasodilation and also reduces systemic blood pressure [11].

In the following test, we used ACh, which is known to promote vasodilation dependent on the presence of the endothelium [12,13]. In pulmonary endothelial cells, the response to this muscarinic agonist showed little calcium mobilization in relation to the other agonists, but with a concentration-dependent profile (Fig. D in supplementary material). The mechanism by which ACh leads to vascular relaxation has been studied and in these studies [12,18], the vasodilating action of endothelium-dependent NO was demonstrated. Atropine acts by blocking all muscarinic receptors (M1 to M5) in a competitive manner, and thus, the actions of ACh and other muscarinic agonists [13]. In pulmonary endothelial cells, it was possible to observe this block in a concentration-dependent manner, but with slight discrimination.

The *C. glaziovii* compounds studied in this work are considered polyphenolic substances with antioxidant activity. These same compounds are present in red wine and some fruits and vegetables. The mechanisms of action by which these compounds perform their activities have not yet been completely clarified, even though they present several activities, such as increased endothelial function, reduced blood pressure [14], inhibition of platelet aggregation, and decreased oxidative activity of low-density lipoproteins (LDL) [15,16].

It has been suggested that these actions may justify the beneficial effect that polyphenolic compounds have on cardiovascular diseases

[17]. Grape extract and red wine are rich in polyphenols. They were studied [19,20] and presented induction of vascular relaxation dependent on the endothelium, possibly through the release of NO or increased activity, increasing the second messenger CGMP.

When evaluating the compounds isovitexin, isoorientin, catechin, epicatechin, procyanidin B2, procyanidin B3, and pure orientin in endothelial cells, we observed the same concentration-dependent pattern for all (Figs. 1–3).

Isovitexin and procyanidin B3, both at concentrations of 100 μM, showed the highest calcium mobilization with levels close to the mobilization seen by KCl, used as a control in these tests. When we associate the antagonist verapamil (Figs. 4–6) with the compounds tested in the endothelial cells, we observed blockade of calcium mobilization induced by the compounds isovitexin, catechin, epicatechin, procyanidins B2, B3, and orientin; while isoorientin mobilized a significant amount of calcium in the highest concentration (100 μM).

In view of these results, it is suggested that the compound isoorientin does not act via L-type calcium channels. The compounds were evaluated together with atropine, in the endothelial cells. All compounds, except isoorientin, act via AT1 by the same route as losartan (Figs. 7–9). The compounds isovitexin, epicatechin, and procyanidin B3 reduced the mobilization of cytosolic calcium while isoorientin, catechin, procyanidin B2, and orientin increased calcium, thus suggesting that metabotropic receptors of the M1 to M5 type are not involved in their effect (Figs. 10–12). The compounds isovitexin, isoorientin, epicatechin, and procyanidin B3 also have an effect via the same route as the HOE 140 antagonist (Figs. 13–15).

5. Conclusion

The results presented showed that in endothelial cells, all compounds, except isoorientin, act by the direct route, similar to verapamil, blocking L-type calcium channels; and the compounds isovitexin, epicatechin, and procyanidin B3 have a mixed effect, also acting through the muscarinic pathway of M1 to M5 metabotropic receptors. All compounds, except isoorientin, act via AT1 by the same pathway as losartan, while the compounds isovitexin, isoorientin, epicatechin, and procyanidin B3 also have an effect via the same pathway as the HOE 140 antagonist.

To our knowledge, the present study shows for the first time the mechanisms of action of compounds derived from *Cecropia glaziouii Sneth* in the endothelial cells, wherein they probably promote their most relevant in vivo beneficial effects. Moreover, we believe that further in-depth research would be interesting to reinforce the mechanisms tested in this work.

In conclusion, from our results, the beneficial effects verified with the popular use of *Cecropia glaziovii* Sneth in pathological conditions, such as systemic arterial hypertension, are consistent with the previously published data about the relaxant and hypotensive effects in rodents treated with Cecropia compounds. Thus, the vasodilating properties of Cecropia can be attributed to the action of flavonoids, flavonoids and procyanidins, through different pathways of calcium mobilization.

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CRediT authorship contribution statement

Gabriella Trettel: Investigation, Methodology, Formal analysis, Writing – original draft. **Clélia Rejane Antonio Bertoncini:** Writing – review & editing. **Maria Teresa Lima-Lindmann:** Conceptualization, Supervision.

Conflict of interest statement

The authors declare that there are no conflicts of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.biopha.2021.112231.

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